

Solution of Tutorial-2
Engineering Mechanics: ME-101 (Semester-II, 2024)
Time: 7:55 – 8:50 AM

Q1. Draw the free-body diagram of the beam as shown in figure 1 that is supported by the pin at B and the roller at A. If the intensity of the distributed load acting on the beam is $w = 6$ kN/m, determine the reactions at roller A and pin B in figure 1. [10 Marks]

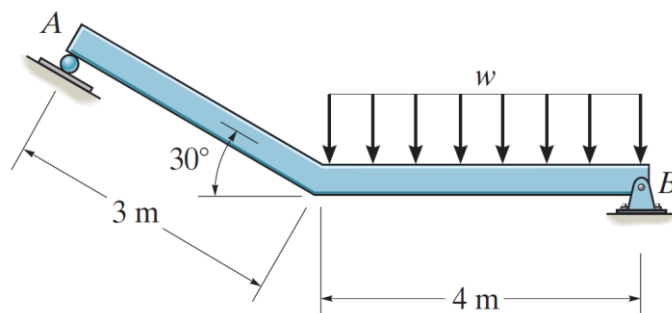


Figure 1

Solution of Q1:

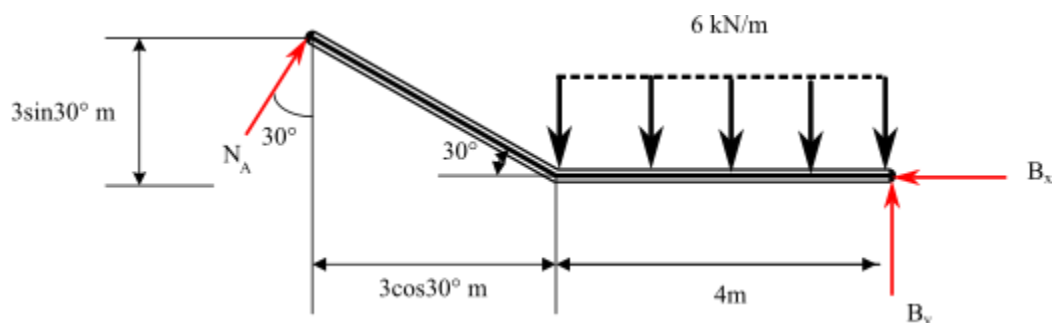


Figure 1.1 **Free-body diagram of the beam** [3 Marks]

Equations of Equilibrium:

The free-body diagram of the problem is shown in Fig.1.1. To calculate the reaction forces at the supports, the distributed load has been replaced with an equivalent concentrated force at the mid-point of the same distributed load as illustrated in Fig. 1.2 below.

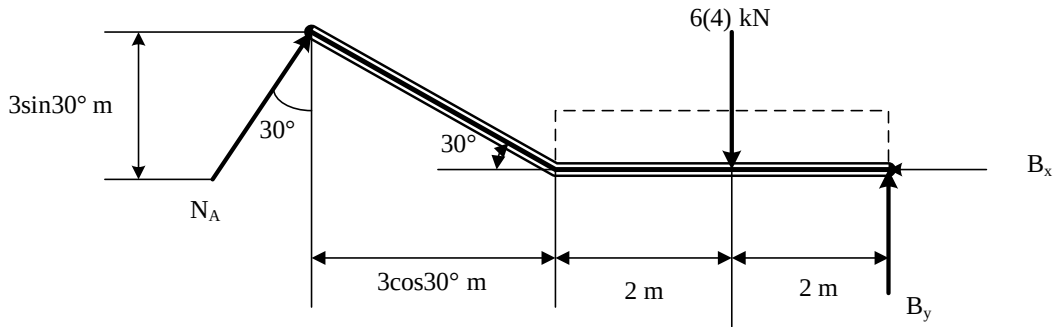


Figure 1.2

Let us consider the counter-clockwise moment as positive and clockwise moment as negative. Now, N_A can be determined directly by writing the moment equilibrium equation about point B by referring to the FBD of the beam as shown in the figure 1.2.

$$\begin{aligned} \sum M_B = 0; \quad & 6(4)(2) - N_A \sin 30^\circ (3 \sin 30^\circ) - N_A \cos 30^\circ (3 \cos 30^\circ + 4) = 0 \\ & N_A = 7.426 \text{ kN} \end{aligned} \quad [3 \text{ Marks}]$$

This result will be used to write the force equilibrium equations along the x and y axes. Let us consider upward and right ward forces as positive.

Now, support reaction at B along the x-direction can be calculated by equating the summations of forces along the x-direction to zero.

$$\begin{aligned} \sum F_x = 0; \quad & 7.426 \sin 30^\circ - B_x = 0 \\ & B_x = 3.713 \text{ kN} \end{aligned} \quad [2 \text{ Marks}]$$

Again, support reaction at B along the y-direction can be calculated by equating the summations of forces along the y-direction to zero.

$$\begin{aligned} \sum F_y = 0; \quad & B_y + 7.426 \cos 30^\circ - 6(4) = 0 \\ & B_y = 17.568 \text{ kN} \end{aligned} \quad [2 \text{ Marks}]$$

2. Determine whether the truss shown in Figure 2 is statically determinate or indeterminate (internally as well as externally). Calculate the force carried by each member using method of joints and show on the figure. All the members are pin connected.

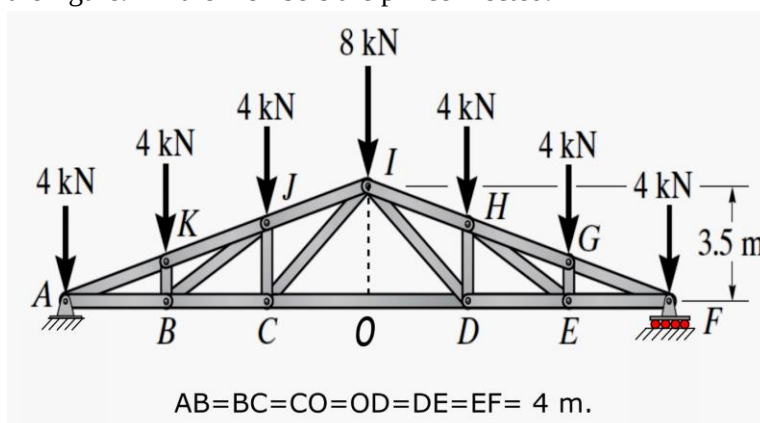
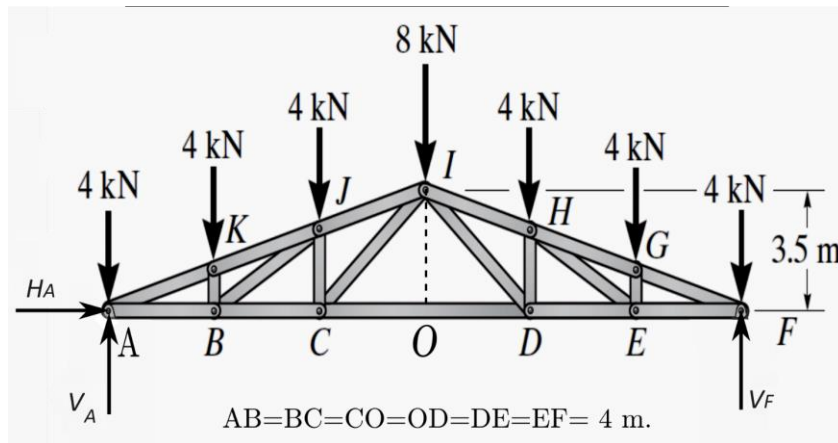


Figure 2

Solution of Q2:



Let the reactions at the hinge support A be H_A and V_A and at the roller support F be V_F . There are total 3 support reactions.

Part 1: Analyzing whether the truss is statically determinate or indeterminate

External indeterminacy $\Rightarrow D_{se} = (\text{No. of external support reactions} - \text{No. of static equilibrium equations available to find them}) = 3 - 3 = 0$ (1 mark)

Internal indeterminacy $\Rightarrow D_{si} = m - (2j - 3)$, where, m = no. of members, j = no. of joints (1 mark)

$$\Rightarrow D_{si} = 19 - (22 - 3) = 0$$

$\Rightarrow$ The truss is statically determinate internally as well as externally.

Part 2: Calculation of the member forces using the method of joints

Tension is indicated by an arrow away from the pin or joint

Compression is indicated by an arrow toward the pin or joint

From the given dimensions of the truss,

$\angle ICO = \tan^{-1}\left(\frac{3.5}{4}\right) = 41.18^\circ$, Now from the concept of similar triangles, $IO/AO = JC/AC = KB/AB$, thus, we can find $JC = 2.33$ m and $KB = 1.167$ m.

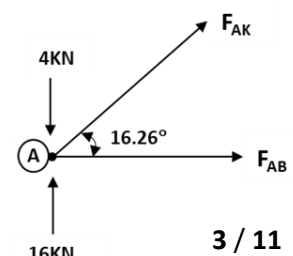
Similarly, $\angle JBC = \tan^{-1}\left(\frac{2.33}{4}\right) = 30.22^\circ$ & $\angle KAB = \tan^{-1}\left(\frac{1.167}{4}\right) = 16.26^\circ$

Now, from the figure it is clear that the truss is **symmetric both with respect to the geometry and the loading**. Hence, the support reactions in the vertical direction $V_A = V_F = 32/2 = 16$ kN

Since there is no horizontal loading, from $\Sigma F_X = 0$, we get, $H_A = 0$. (1 mark)

Now solving for the unknown member forces with the help of FBD at the joints –

At joint A, $\Sigma F_Y = 0$ (considering upward forces positive)



$\Rightarrow F_{AK} \sin 16.26^\circ + 16 - 4 = 0 \Rightarrow F_{AK} = -42.86 \text{ kN}$ (negative sign indicates that the direction of the force F_{AK} is opposite to what has been assumed in the FBD; which implies that it is compressive in nature)

$\Sigma F_X = 0$ (considering left to right forces positive)

$$\Rightarrow F_{AB} + F_{AK} \cos 16.26^\circ = 0 \Rightarrow F_{AB} = -F_{AK} \cos 16.26^\circ = 42.86 \cos 16.26^\circ = 41.15 \text{ kN}$$

$\therefore F_{AB} = 41.15 \text{ kN}$ (Tensile) & $F_{AK} = 42.86 \text{ kN}$ (Compressive)

(1 marks)

At joint K, $\Sigma F_X = 0$ (considering left to right forces positive)

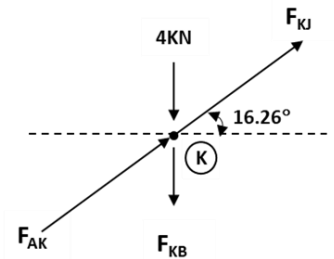
$$\Rightarrow F_{KJ} \cos 16.26^\circ + F_{AK} \cos 16.26^\circ = 0 \Rightarrow F_{KJ} = -F_{AK} = -42.86 \text{ kN}$$

$\Sigma F_Y = 0$ (considering upward forces positive)

$$\Rightarrow F_{KJ} \sin 16.26^\circ + F_{AK} \sin 16.26^\circ - F_{KB} - 4 = 0 \Rightarrow F_{KB} = -4 \text{ kN}$$

$\therefore F_{KB} = 4 \text{ kN}$ (Compressive) & $F_{KJ} = 42.86 \text{ kN}$ (Compressive)

(1 marks)



At joint B, $\Sigma F_Y = 0$ (considering upward forces positive)

$$\Rightarrow F_{BJ} \sin 30.22^\circ - F_{KB} = 0 \Rightarrow F_{BJ} = 7.95 \text{ kN}$$

$\Sigma F_X = 0$ (considering left to right forces positive)

$$\Rightarrow F_{BJ} \cos 30.22^\circ + F_{BC} - F_{AB} = 0 \Rightarrow F_{BC} = 34.28 \text{ kN}$$

$\therefore F_{BJ} = 7.95 \text{ kN}$ (Tensile) & $F_{BC} = 34.28 \text{ kN}$ (Tensile)

(1 marks)

At joint J, $\Sigma F_X = 0$ (considering left to right forces positive)

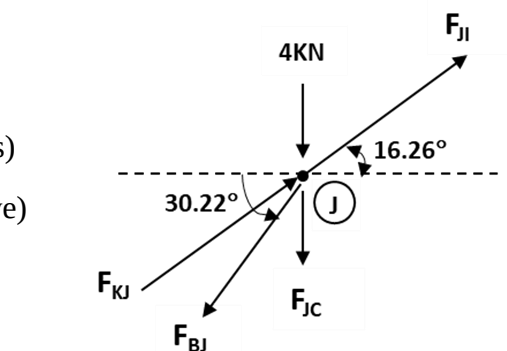
$$\Rightarrow F_{JI} \cos 16.26^\circ + F_{KJ} \cos 16.26^\circ - F_{BJ} \cos 30.22^\circ = 0$$

$$\Rightarrow F_{JI} = -35.70 \text{ kN}$$

$\Sigma F_Y = 0$ (considering upward forces positive)

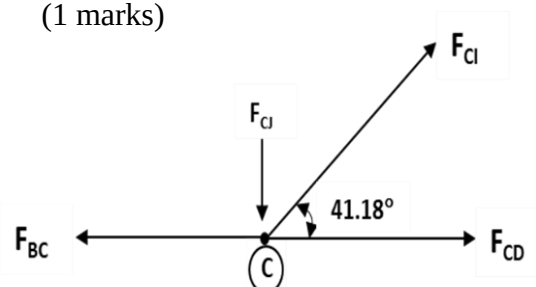
$$\Rightarrow F_{JI} \sin 16.26^\circ + F_{KJ} \sin 16.26^\circ - F_{BJ} \sin 30.22^\circ - F_{JC} - 4 = 0 \Rightarrow F_{JC} = -6 \text{ kN}$$

$\therefore F_{JI} = 35.70 \text{ kN}$ (Compressive) & $F_{JC} = 6 \text{ kN}$ (Compressive)



At joint C, $\Sigma F_Y = 0$ (considering upward forces positive)

$$\Rightarrow F_{CI} \sin 41.18^\circ - F_{JC} = 0 \Rightarrow F_{CI} = 9.11 \text{ kN}$$



$\Sigma F_x = 0$ (considering left to right forces positive)

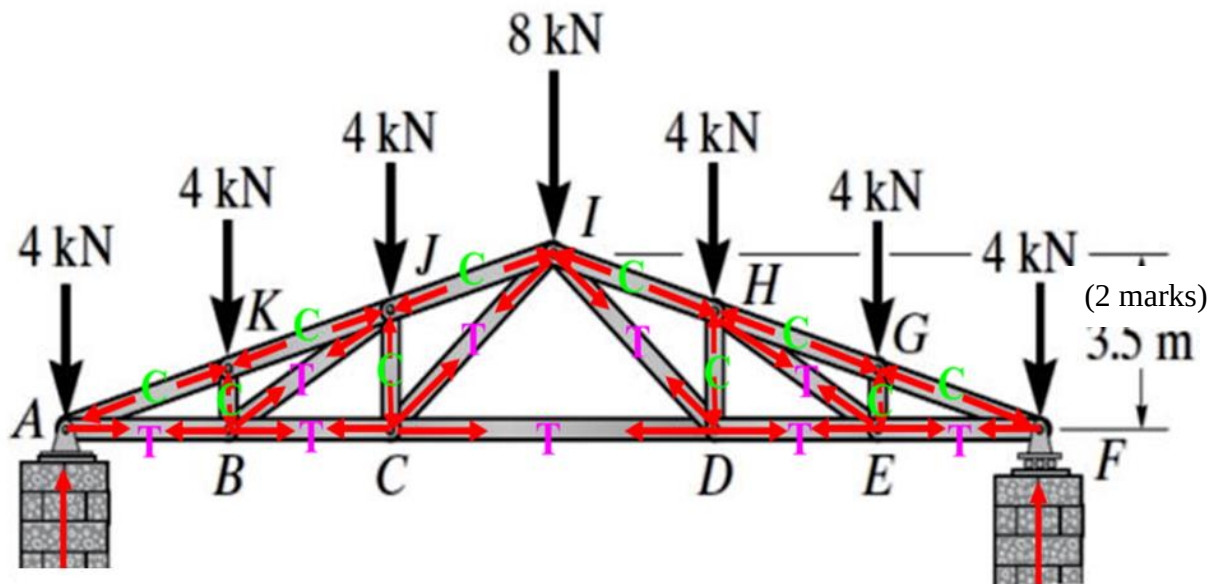
$$\Rightarrow F_{CI} \cos 41.18^\circ + F_{CD} - F_{BC} = 0 \Rightarrow F_{CD} = 27.42 \text{ kN}$$

$\therefore F_{CI} = 9.11 \text{ kN}$ (Tensile) & $F_{CD} = 27.42 \text{ kN}$ (Tensile) (1 marks)

Since the truss is symmetric with respect to both geometry and loading, the other half portion of the truss will have the same member force values as calculated above. Thus, the reaction forces and the member forces are as follows: $V_A = V_F = 16 \text{ kN}$ (upward); and $H_A = 0$

$F_{AB} = F_{FE} = 41.15 \text{ kN}$ (Tensile); $F_{BC} = F_{ED} = 34.28 \text{ kN}$ (Tensile); $F_{CD} = 27.42 \text{ kN}$ (Tensile); $F_{AK} = F_{FG} = 42.86 \text{ kN}$ (Compressive); $F_{KJ} = F_{GH} = 42.86 \text{ kN}$ (Compressive); $F_{JI} = F_{HI} = 35.70 \text{ kN}$ (Compressive); $F_{KB} = F_{EG} = 4 \text{ kN}$ (Compressive); $F_{JC} = F_{DH} = 6 \text{ kN}$ (Compressive); $F_{BJ} = F_{EH} = 7.95 \text{ kN}$ (Tensile); $F_{CI} = F_{DI} = 9.11 \text{ kN}$ (Tensile)

The direction of all the member forces and support reactions of the given truss are as follows-



(2 marks)

Q3. For the plane truss shown in Figure 3, find out the forces in members in FE , FC , and BC by considering all members as pin connected using method of sections.

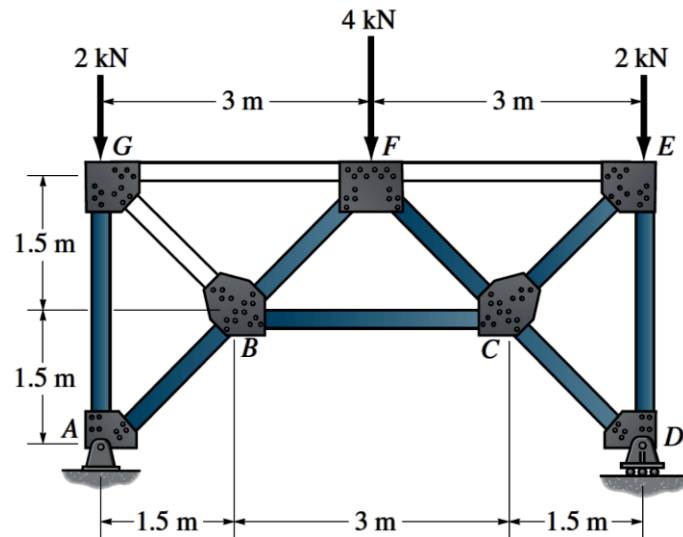


Figure 3

Solution of Q3:

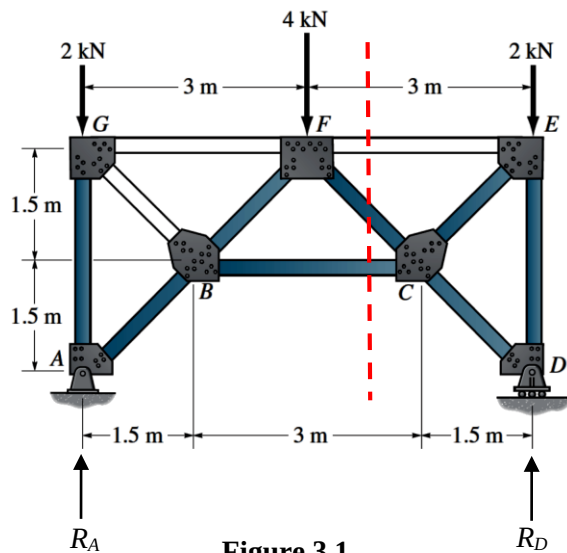


Figure 3.1

(1 mark)

Let the reactions at A and D be R_A and R_D respectively. (Note that the horizontal reaction at A is zero)

Total sum of vertical forces, $\sum V = 0$

$$\Rightarrow R_A + R_D = 2 + 4 + 2 = 8 \text{ kN}$$

Moment about point A, $\sum M_A = 0$

$$\Rightarrow R_D \times 6 - 2 \times 6 - 4 \times 3 = 0$$

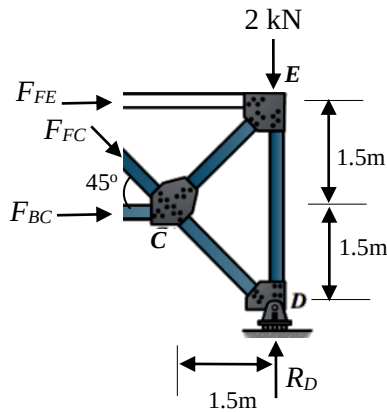
$$\Rightarrow R_D = 4 \text{ kN}$$

(1 mark)

Now, as $R_A + R_D = 8 \text{ kN}$, therefore, $R_A = 4 \text{ kN}$

(1 mark)

Now, for the member forces by method of sections, a section has been cut as shown in the Figure 3.1, and the resultant force diagram in shown in Figure 3.2.



(2 marks)

Figure 3.2

Moment about C due to all the member forces in the cut section and the external forces should be zero. Assuming Counter-clockwise moments to be positive:

$$\begin{aligned}\sum M_C &= 0 \\ \Rightarrow R_D \times 1.5 - 2 \times 1.5 - F_{FE} \times 1.5 &= 0 \\ \Rightarrow F_{FE} &= 2 \text{ kN (Compression)}\end{aligned}$$

(2 marks)

Now, sum of horizontal forces should equate to zero

$$\begin{aligned}\sum H &= 0 \\ \Rightarrow F_{BC} + F_{FE} + F_{FC} \cos 45^\circ &= 0 \\ \Rightarrow F_{BC} + 2 + \frac{F_{FC}}{\sqrt{2}} &= 0 \quad \text{----- (1)}\end{aligned}$$

Again, sum of vertical forces should equate to zero

$$\begin{aligned}\sum V &= 0 \\ \Rightarrow R_D - 2 - F_{FC} \sin 45^\circ &= 0 \\ \Rightarrow 4 - 2 - \frac{F_{FC}}{\sqrt{2}} &= 0 \quad \{ \text{as } R_D = 4 \} \\ \Rightarrow F_{FC} &= 2\sqrt{2} \text{ kN} = 2.83 \text{ kN (Compression)}\end{aligned}$$

(2 marks)

Now, from equation (1),

$$F_{BC} = -4 \text{ kN (Tension)}$$

(1 mark)

Q4. As shown in **Figure 4**, the members CJ and CF of the loaded truss cross but are not connected to members BI and DG, respectively. Compute the forces in members BC, CJ, CI, and HI.

[10 marks]

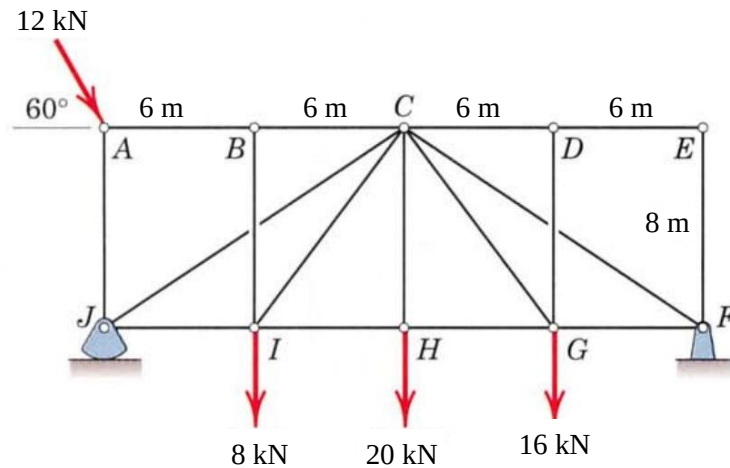


Figure 4

Solution of Q4:

Equilibrium Equations:

Let us consider the counter-clockwise moment as positive and clockwise moment as negative. From Fig. 4, considering the equilibrium of the whole truss and taking moment about point F,

$$\sum M_F = 0$$

$$-(12 \cos 60^\circ) \times 8 - R_J \times 24 + (12 \sin 60^\circ) \times 24 + 8 \times 18 + 20 \times 12 + 16 \times 6 = 0$$

$$R_J = 28.4 \text{ kN} \dots \dots \dots (2 \text{ marks})$$

Using method of section:

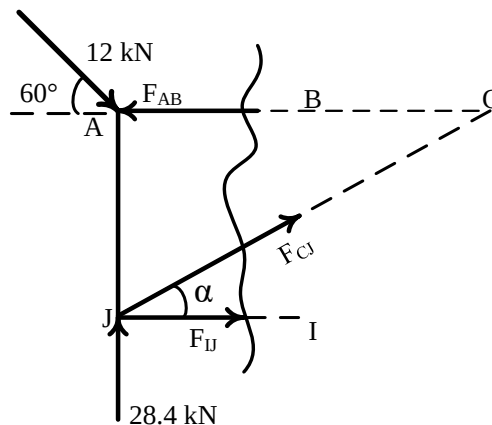


Fig. 4.1

Let us consider upward and right ward forces as positive. Now, from Fig. 4.1, force in truss member CJ can be calculated by equating the summations of forces along the y-direction.

$$\sum F_y = 0, \rightarrow 28.4 - 12 \sin 60^\circ + F_{CJ} \sin \alpha = 0, \text{ where, } \alpha = \tan^{-1} \left(\frac{8}{12} \right) = 33.7^\circ$$

,

$$\therefore F_{CJ} = -32.45 \text{ kN (Compressive)} \dots \dots \dots (2 \text{ marks})$$

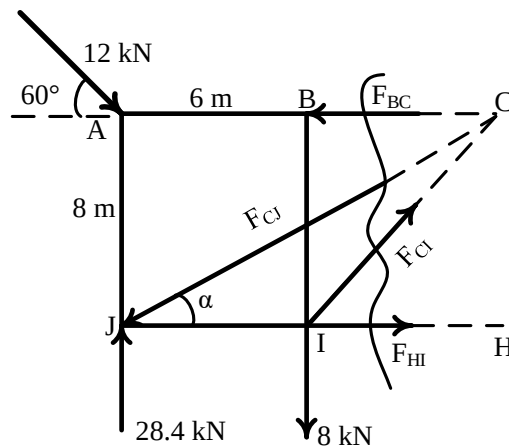


Fig. 4.2

Now, from Fig. 4.2, force in truss member CI can be calculated by equating the summations of forces along the y-direction.

$$\sum F_y = 0$$

$$-12 \times \sin 60^\circ + 28.4 - 8 - 32.45 \times \sin(33.37^\circ) + F_{CI} \times \left(\frac{8}{10}\right) = 0$$

$$\therefore F_{CI} = 10 \text{ kN (Tensile) (2 marks)}$$

Let us consider the counter-clockwise moment as positive and clockwise moment as negative and taking moment about point C.

$$\sum M_C = 0$$

$$(12 \sin 60^\circ) \times 12 - (28.4) \times 12 + 8 \times (6) + F_{HI} \times (8) = 0$$

$$\therefore F_{HI} = 21 \text{ kN (Tensile) (2 marks)}$$

Now, from Fig. 4.2, force in truss member BC can be calculated by equating the summations of forces along the x-direction.

$$\sum F_x = 0$$

$$12 \times \cos 60^\circ - 32.45 \times \cos 33.37^\circ + 10 \times \left(\frac{6}{10}\right) + 21 - F_{BC} = 0$$

$$F_{BC} = 6 \text{ kN (Compressive) (2 marks)}$$

Q5. A vertical force P on the foot pedal of the bell crank is required to produce a tension T of 200 N in the vertical control rod. Draw FBD and determine the corresponding bearing reactions at A and B as shown in **Figure 5**.

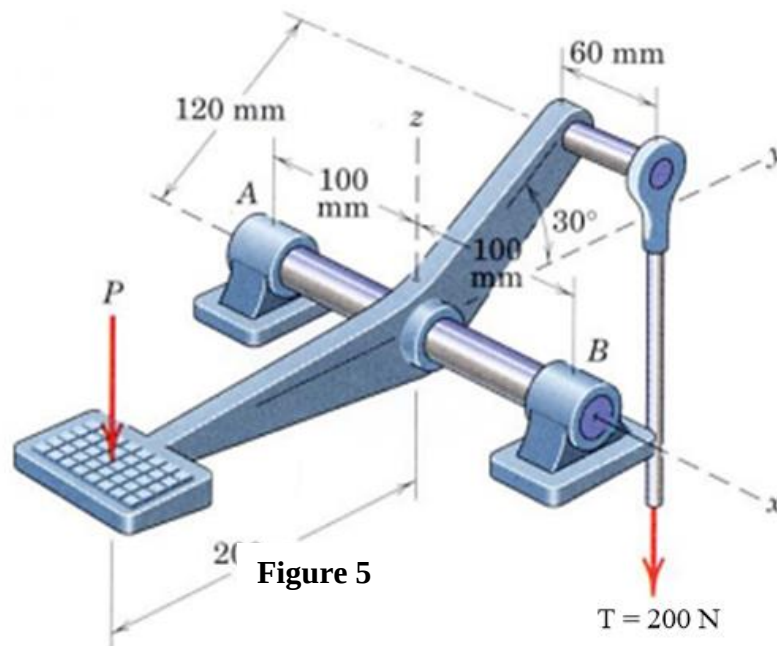


Figure 5

Solution of Q5:

If we look along y -axis, the free-body diagram appears to be as in Fig. 5.1.

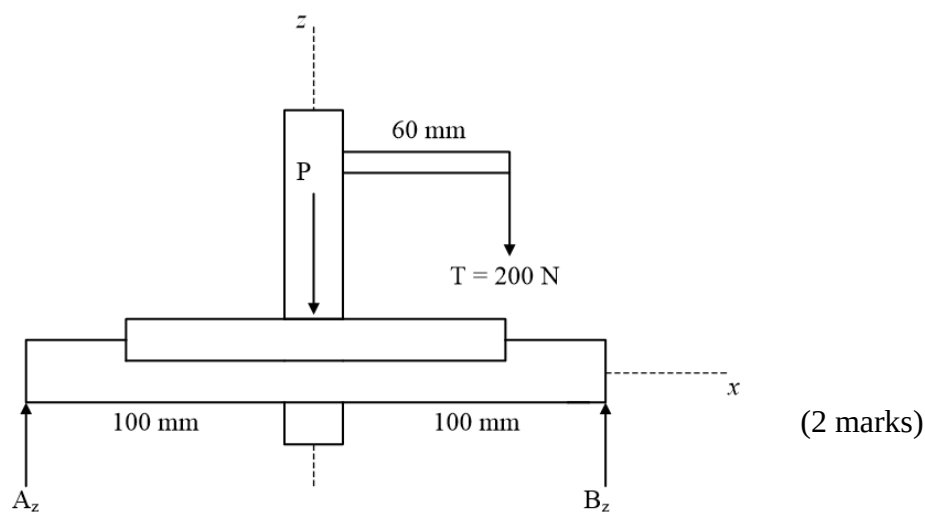


Figure 5.1

Considering the summation of moments about x axis equals zero, to find the vertical force P from Figure 6, we have:

$$\curvearrowleft + \sum M_x = 0: \quad P(200) - 200(120 \cos 30^\circ) = 0$$

$$P = 103.92 \text{ N}$$

(2 marks)

To find the bearing reaction at B, summation of moments about y axis is equated to zero, we have:

$$\curvearrowright \sum M_y = 0, \rightarrow A_z(100) - B_z(100) + 200 \times 60 = 0$$

$$A_z - B_z = -120 \dots\dots(5a) \quad (1 \text{ marks})$$

Now, considering the summation of all the forces along z-direction equals zero from figure 5.1, we have:

$$\sum F_z = 0, \rightarrow A_z + B_z - 200 - P = 0 \quad (1 \text{ marks})$$

$$A_z + B_z = 303.92 \dots\dots\dots(5b)$$

$$\text{Solving Eqns. (5a) and (5b) one gets, } A_z = 91.96 \text{ N, } B_z = 211.96 \text{ N} \quad (2 \text{ marks})$$

As, there are no external forces along x and y-directions, so, the corresponding bearing reactions are:

$$A_x = B_x = 0 \quad (2 \text{ marks})$$

$$A_y = B_y = 0$$